Jerome Pitogo de Leon, Ph.D.

Postdoctoral researcher at The University of Tokyo

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Overview

Research: Author of 103 refereed publications in exoplanet science with 2,400 total citations and an h-index of 30 (as of 2026-10-17).[1] Research focuses on the discovery and characterization of transiting exoplanets using space- and ground-based telescopes and advanced data analyses.

Teaching: Designed and delivered introductory astronomy courses for undergraduates, five of whom subsequently pursued graduate studies in the Philippines and abroad.

Mentorship: Supervised 2 PhD, 2 master's, and 4 undergraduate students on astronomy research projects. Developed publicly available software tools to support research and instruction.

Outreach: Delivered invited talks for academic and public audiences and founded an astronomy community initiative to promote engagement in astronomy.

Education

PhD in Astronomy, 2021
Department of Astronomy, The University of Tokyo, Japan
Dissertation: *Discovery and Characterization of Transiting Exoplanets with Diverse Radii and Ages* (online)
Advisor: Prof. Motohide Tamura

MSc in Astronomy, 2018
Department of Astronomy, The University of Tokyo, Japan
Thesis: *Multi-color Simultaneous Transit Observations of Low-density Hot Jupiters* (online)
Advisor: Prof. Motohide Tamura

BSc in Physics, 2013
Department of Physics, University of the Philippines Los Baños, Laguna, Philippines
Thesis: *Thermal Differential and Image Analysis of a Fabricated 11-cm Solar Telescope Using an Active-Passive Cooling System*
Advisor: Dr. Rogel Mari Sese

Full-time Work Experience

Project Assistant Professor (2025/4–Present)
Komaba Institute for Science, Graduate School of Arts and Sciences, The University of Tokyo, Japan
Role: Lead coordination of observations and data analysis with MuSCAT4 under the JSPS Kiban S project, including planning observing campaigns, overseeing data reduction, and

[1] Publication metrics from NASA ADS.

maintaining analysis workflows.

Project Research Assistant (2023/4–2025/3)
Department of Multi-Disciplinary Sciences, Graduate School of Arts and Sciences, The University of Tokyo, Japan
Role: Led the discovery and detailed characterization of young transiting exoplanets using TESS photometry, including target vetting, light-curve modeling, and follow-up coordination.
Advisor: Prof. Norio Narita (see de Leon et al. 2023)

Project Research Assistant (2021/4–2023/3)
Department of Astronomy, Graduate School of Science, The University of Tokyo, Japan
Role: Coordinated an international collaboration that identified 37 transiting exoplanets from Kepler data, performing transit detection, validation, and system characterization.
Advisor: Prof. Motohide Tamura (see de Leon et al. 2021)

Research Development Assistant (2013/4–2014/6)
Regulus SpaceTech, Inc., Los Baños, Laguna, Philippines
Role: Initiated and implemented software for automated solar telescope control and data analysis.
Advisor: Dr. Rogel Sese

Temporary Positions

Consultant (2024/6–Present)
Telos Media (www.telos.so)
Role: Accuracy reviewer of Astronomy-related long-form educational videos published on youtube.

Remote Lecturer (2021/6–2021/10)
Faculty of Aerospace Engineering, School of Engineering & Architecture, Ateneo de Davao University, Philippines
Role: Designed and delivered lectures on introductory astronomy and astronomical data analysis for senior undergraduates.

Technical Assistant (2018/10–2019/3)
Engineering Department, The University of Tokyo, Japan
Role: Designed and implemented image-processing software for stitching featureless high-resolution images.
Advisor: Assoc. Prof. Tomohiko Hayakawa (see Murakami et al. 2018)

Research Assistant (2017/4–2018/3)
Department of Astronomy, The University of Tokyo, Japan
Role: Designed a pipeline for modeling and analyzing multi-wavelength transit light curves and mentored new group members in its use.

Research Student (2015/9–2016/3)
Department of Astronomy, Graduate University for Advanced Studies (SOKENDAI), Japan
Role: Conducted data reduction and analysis of high-resolution astronomical images and prepared technical content for scientific reports.
Advisor: Dr. Satoshi Mayama

Intern (2014/7–2014/9)
Academia Sinica Institute for Astronomy and Astrophysics (ASIAA), Taipei, Taiwan

Role: Developed an image-processing pipeline to analyze high-resolution images of a young forming star using the 8-m Subaru Telescope.
Advisor: Dr. Michihiro Takami (see de Leon et al. 2015)

Skills

Programming & Scientific Computing: Python (NumPy, SciPy, Pandas, JAX, NumPyro); experience in building complex data analysis pipelines for large temporal and spatial datasets

Data Analysis & Modeling: Time-series analysis, Bayesian inference, probabilistic modeling, optimization, statistical analysis of large survey data, technical writing, project management

Software & Systems: Linux/Unix environments, server administration (Grafana monitoring), Bash scripting, version control (Git), automation, reproducible research workflows

Visualization & Communication: Matplotlib, scientific visualization, academic writing, conference presentations, public outreach

Leadership & Mentorship: Research group supervision, student advising (PhD, MSc, undergraduate), curriculum design, international collaboration

Languages: Filipino (native), English (fluent), Japanese (working proficiency)

Awards

2022/12: St. Joseph's Academy Centennial Achiever's Award

2021/3: Philippine Embassy in Japan Medal for Distinguished Filipino Graduates

2018/4–2021/3: MEXT PhD Scholarship, Ministry of Education, Culture, Sports, Science and Technology (JPY 8.4M)

2016/4–2018/3: MEXT MSc Scholarship (JPY 5.6M)

2015/4–2016/3: MEXT Japanese Language and Research Scholarship (JPY 2.8M)

2015/10–2016/3: Research Grant, Graduate University for Advanced Studies (JPY 0.2M)

2014/9: SanDisk Japan Scholarship (USD 3.75k, deferred)

2013/3: International Space University Scholarship (EUR 9k, deferred)

2011–present: Multiple travel grants for data science workshops and research conferences, including observation runs in the U.S., Europe, South Africa, and Southeast Asia

Teaching and Mentorships

Designed and Taught Courses: Created and delivered two elective courses for 4th-year Aerospace Engineering undergraduates at the Ateneo de Davao University during the 2nd semester of 2021.

Research Collaborations: Collaborated with several international teams on multi-year projects resulting in research publications in high-impact, peer-reviewed journals.

Pipeline Development: Developed publicly available Python code to analyze data from the TESS telescope for research and instructional use (see github.com/jpdeleon/quicklook).

Undergraduate Thesis Advisory: Mentored K.M. Shariat on Bayesian Statistical Modeling; mentored K. Uno on estimating stellar ages using multiple techniques as part of a Research Introduction class; mentored H. Kobayashi on developing a pipeline to find young transiting planets and binary stars in nearby clusters (see github.com/hiremasa/ytps). Also advised A.

Nodado on an undergraduate thesis on variable star characterization.

MEXT Scholarship Guidance: Guided A. Javier and C. Cabatlao from application to acceptance, supporting their successful pursuit of graduate studies in Japan.

Trainings and Workshops

Nature Masterclasses: Impressive Academic Presentations (Expert trainer-led full-day workshop), University of Tokyo, March 18, 2026

University of Tokyo Faculty Development (UTokyo FD Open) Program, Tokyo, Japan, Oct 2025–Feb 2026

5th ExoClock Meeting, Madrid, Spain, Oct 4–6, 2025

ARIEL Consortium Meeting, Madrid, Spain, Sep 30–Oct 4, 2025

Young Transiting Planet Workshop 2025, Ishigaki, Japan, Mar 12–15, 2025

JWST Data Analysis Workshop, Chiang Mai, Thailand, Jun 24–Jul 5, 2024

Japan Planetary Atmosphere Research Society Workshop II, Atami, Japan, Mar 2023

Japan Planetary Atmosphere Research Society Workshop, Matsushima, Miyagi, Japan, Mar 2022

RESCEU Symposium: From Protoplanetary Disks through Planetary System Architecture to Planetary Atmospheres and Habitability, Okinawa, Japan, Oct 14–18, 2019

1st NARIT–EACOA Summer Workshop on Astrostatistics and Informatics, Chiang Mai, Thailand, Aug 13–17, 2019

Telluric Line Hack Week, Flatiron Institute, New York, USA, Feb 25–28, 2019

NASA Sagan Exoplanet Summer Workshop, NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, USA, Jul 23–27, 2018

Japan Geoscience Union Meeting, Chiba, Japan, May 20–24, 2018

2nd Rencontres du Vietnam on Exoplanetary Science, Binh Dinh, Vietnam, Feb 2018

TIARA Summer School on Astrostatistics and Big Data, Taipei, Taiwan, Sep 4–8, 2017

Astronomy Hack Week, University of Washington, Seattle, WA, USA, Aug 28–Sep 1, 2017

Protoplanetary Disk Dynamics and Planet Formation Workshop, Yokohama Institute for Earth Sciences (JAMSTEC), Kanagawa, Japan, Sep 29–Oct 2, 2015

27th Spring School in Particles and Fields, Chang Gung University, Taoyuan City, Taiwan, Apr 2–5, 2014

NARIT–KASI Radio Astronomy Winter School, Chiang Mai, Thailand, Jan 27–30, 2014

TEDx Diliman: Things That Matter, UP Diliman, Philippines, Oct 20, 2013

2nd Southeast Asian Astronomers Collaboration Meeting, Bandung, Indonesia, Nov 18–22, 2013

Sokendai Asian Winter School on Exoplanets, National Astronomical Observatory of Japan (NAOJ), Mitaka, Tokyo, Japan, Dec 3–5, 2012

1st Southeast Asian Astronomers Collaboration Meeting, Palawan, Philippines, Nov 5–7, 2012

Astronomy Training Workshop for Science Educators, Galileo Teacher Training Program (GTTP), UP Los Baños, Philippines, Apr 9–11, 2012

The Universe Survives 2012, NIDO Science Discovery Center, SM Mall of Asia, Pasay City, Philippines, Feb 18 and 24, 2012

Search for Undiscovered Main Belt Asteroids and Other Important NEO Observations,

International Asteroid Campaign (IASC), Philippines, Apr 20–May 20, 2012

29th Samahang Pisika ng Pilipinas (Physics Society of the Philippines) National Physics Congress, National Institute of Physics, College of Science, UP Diliman, Philippines, Oct 24–26, 2011

International Astronomical Union (IAU) Gala Night, NIDO Discovery Center, SM Mall of Asia, Pasay City, Philippines, Oct 21, 2011

Developing Astronomy Education and Research in the Philippines, UP Los Baños, Philippines, Oct 20, 2011

Outreach Activities

Resource speaker for Junior High School Career Week, SJASQ, Philippines, Jan 24, 2026

Guest on a podcast titled "Are We Alone?" (published on Spotify), Dec 2025

Resource speaker for Philippine Astronomy Week 2024 led by PAGASA, Philippines, Feb 2024

Guest at Pinoy Scientists "Pathways to Science" (Aug 2023)

Featured by Pinoy Scientists (Jan 2023)

Resource speaker for Science Week at De La Salle Integrated School, Philippines, Mar 2022

Resource speaker for the Philippine Science High School–Western Visayas Campus, Philippines, Mar 2021

Courtesy call with the Director of the Philippine Space Agency (PhilSA), Mar 2021

Invited speaker at the National Institute of Physics, University of the Philippines Diliman (UPD), Jan 2020

Astronomy Over Coffee (self-funded astronomy event), UP Diliman Hotel, Sep 2019

Founder of the Filipino Astronomy Community (FAC; 2016)

Invited speaker at Philippine Rural High School, Los Baños, Philippines, Apr 2014

Invited speaker at St. Joseph's Academy Sariaya (SJASQ) High School, Philippines, Apr 2013

Affiliations and Services

Monthly Notices of the Royal Astronomical Society (MNRAS; Impact Factor: 4.8; active referee)

SPACETIDE Foundation (spacetide.jp; active intern, 2022–present)

Science Integrated Direction for High School Investigators (SIDHI; mentor, 2022–present)

Science and Technology Advisory Council–Japan Chapter (STAC-J; member, 2019–present)

Exoplanet Follow-up Observing Program (ExoFOP; member, 2018–present)

Association of Filipino Students in Japan (AFSJ; president, 2017–2018)

Association of Filipino Scholars in Taiwan (AFST; founding member, 2014)

Gawad Kalinga (volunteer, 2009–2013)